

AN IMPROVED PARALLEL SIMULATED ANNEALING ALGORITHM USED FOR PROTEIN STRUCTURE PREDICTION

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Abstract:

This paper introduces an improved Simulated Annealing --Parallel Simulated Annealing with Genetic Crossover: PSAGC, and uses this method in energy minimization problem of protein structure prediction. Through experiments on three real proteins, PSAGC is proved more effective than SA and PSA, it can achieve conformations which have lower energy values. Then the paper investigates crossover interval and crossover method of PSAGC, and obtains useful conclusions.

Keywords:

Protein Structure Prediction; Energy Minimization; Simulated Annealing; Crossover Interval

1. Introduction

In Recent years, the protein structure prediction problem has been focused much attention upon. The design of new medicine, creating an artificial protein and analysis of genetic disease are achieved from the resolution of the protein structure prediction problem [1, 2]. The protein is a natural substance and an essential part in the organic phenomenon, and its biological function is specified by the 3-D structure. Although the structures of proteins can be accurately determined by the experimental techniques of NMR (Nuclear Magnetic Resonance) and X-ray crystallography [2, 3], but there are also thousands more to be discovered, each with a unique structure and special properties. Therefore, there is a great interest in develop in computational approaches to correctly predict the 3D structure of proteins.

Anfinsen's thermodynamic hypothesis states that the amino acid sequence alone provides adequate information for finding the native conformation of a protein, since a protein in its surrounding environment attempts to minimize the free energy of the system [4]. Therefore, the protein exists at the global minimum free energy state given a set of environmental conditions.

The natural conformation of the protein in real world is corresponded to free-energy minimal state and it is predictable by computational search with optimization algorithm. SA (Simulated Annealing) has often been employed as the optimization method to predict the tertiary structure of proteins by the minimization of the energy [5, 6].

In this study, we apply a hybrid algorithm of using SA with GA (Genetic Algorithm) operation. SA is good at searching locally and GA is good at searching globally [7]. The proposed algorithm is called PSAGC (Parallel Simulated Annealing using Genetic Crossover) [8, 9]. In this algorithm, plural processes of SA are running separately. GA's crossover is performed in exchange of the transient solution that has been found by each SA.

2. Parallel Simulated Annealing using Genetic Crossover (PSAGC)

2.1. Overview of PSAGC

PSAGC is the optimization method. In this algorithm, several SA processes are running in parallel. After some steps, these processes are stopped and the temporal solutions among SAs running in parallel by the genetic crossover, which is one of the GA operator [7]. In case one SA obtains an optimum value of one of the design variables, the crossover operation works to transfer the value to the other SA. It is expected that the optimum solution can be derived in the early stage. In this algorithm, the total number of SAs running in parallel is defined as population size and the solution of each SA is corresponded to one individual of GA. The following steps are the flow of PSAGC:

Step 1 Assume an initial temperature. Initial individuals are generated randomly.

Step 2 Each individual is given to one SA as an initial searching point. SA is executed in parallel in the order of

generating a new solution. Then accept the solution (In this paper, the Metropolis standard is used as the acceptance criterion), and cooling a temperature.

Step 3 When annealing has been done for the specified steps (CI, Crossover Interval), The pairs are generated randomly from those solutions. The number of pairs is a half number of the individuals.

Step 4 The crossover is applied to a pair and two children are generated.

Step 5 The best two individuals which have higher fitness values, lower energy values are selected from the four individuals (two parents and two children). These two individuals are new initial searching points for the next SA.

Step 6 The operations from Step 4 to Step 5 are applied to all the pairs.

Step 7 The steps between two to six are iterated until the terminal criterion is satisfied.

2.2. Applying PSAGC to energy minimization

The target proteins in this paper is Met-enkephalin, C-peptide and Parathyroid Hormone Fragment: PTH(1-34). Met-enkephalin is a small protein that consists of only five amino acids. C-peptide and PTH(1-34) are larger than Met-enkephalin. These proteins are described in table 1.

Table 1. Description of Proteins

Protein	Amino Acids	Dihedral Angles	Atom Number
Met-enkephalin	5	19	75
C-peptide	13	64	218
PTH (1-34)	34	178	582

The dihedral angles of backbone and side chains are applied as design variables. In other words, each individual in the algorithm is represented by the binary codes of the dihedral angles. Values of the dihedral angles are in the range of $[-180^\circ, 180^\circ]$. Each dihedral angle is generated and given the accept criterion sequentially, and then the temperature is cooled. This series of operation is defined as a MCsweep (Monte Carlo Sweep) [10]. Parameters and geometric information of ECEPP/2 (Empirical Conformational Energy Program for Peptides and Proteins) [11-13] are used for the energy function.

3. Comparison with other methods

In the experiments, we compare the searching ability of PSAGC, SA and PSA for minimizing the energy of protein tertiary structures, Parameters of the three

algorithms are shown in table 2. The comparison is shown in table 3.

Table 2. Parameters of Algorithms

Parameter	PSAGC	PSA	SA
Initial Temperature	2.0	2.0	2.0
Terminal Temperature	0.1	0.1	0.1
CI	16 Csweep	-	-
Parallel Number	8	8	-

Table 3. Comparison of Searching Capability

	Met-enkephalin (kcal/mol)	C-peptide (kcal/mol)	PTH(1-34) (kcal/mol)	Met-enkephalin success rate
10	-11.0	-42.0	-210.0	0.62
PSA	-11.5	-38.9	-229.0	0.88
PSAGC	-11.1	-53.9	-246.0	0.90

The results in table 3 describes that to larger proteins C-peptide and PTH(1-34), PSAGC can receive the conformations which have lower energy than SA and PSA's.

To small protein Met-enkephalin, the three algorithms can all receive the conformations which have low energy (It has been already derived that the energy of this protein is less than -11kcal/mol [9]. In this paper, it is also considered that the optimum structure is derived when the structure has the energy less than -11kcal/mol). Then, to Met-enkephalin, we compare the success rate of receiving the optimum structures of these algorithms. The last column of table 3 shows the result. Obviously, the success rate of PSAGC is higher than the other two algorithms.

In a word, the searching ability of PSAGC is better than SA and PSA.

4. Crossover Interval of PSAGC

When applying GA crossover operator, we consider different CI, 16Mcsweep, 32Mcsweep, 64Mcsweep. In Figure 1-3, the energy transitions of different CI are illustrated.

From Figure 1, 2, we can know that, to larger proteins like C-peptide and PTH(1-34), CI 16 is the best, to small protein Met-enkephalin, CI 64 is the best. Such result is because Met-enkephalin is a small protein which has only 19 dihedral angles, accordingly, the energy minimization problem should be simple. If the CI is very short, the variety of individuals will be lost, and the performance of the algorithm will be not so good.

In conclusion, to small proteins, the affection of CI is not obvious. Indeed, frequent crossover will bring about bad influence. But to bigger proteins, crossover operator can improve the performance of the algorithm, and the shorter the CI is, the better the result will be.

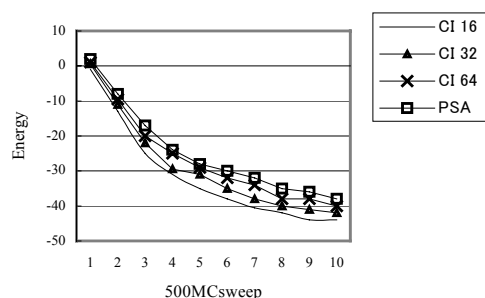


Figure 1. Energy transition of C-peptide

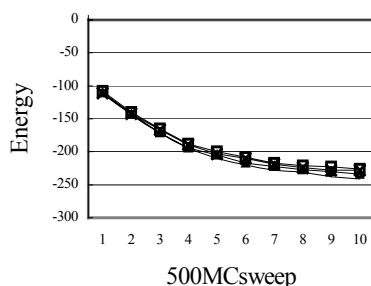


Figure 2. Energy Transition of PTH (1-34)

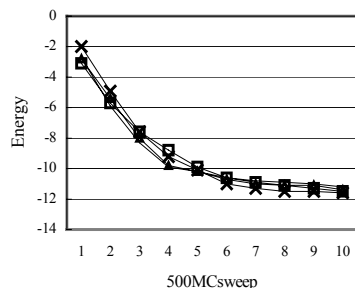


Figure 3. Energy Transition of Met-enkephalin

5. Crossover Methods

In this chapter, C-peptide is the target protein,

- ◇ CI : 16MCsweep
- ◇ Parallel Number : 8
- ◇ 6000MCsweep

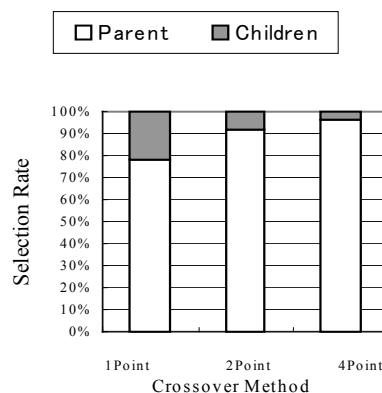


Figure 4. Selection Rate of Parents and Children

Figure 4 shows the selection rate of parents and children after genetic crossover, obviously, one point crossover method has the highest children selection rate, approximately 20%, two point and four point crossover are both less than 10%. Obviously most of the children after crossover can not be selected. It implies the crossover is not effective enough. So how to make the genetic crossover more effective is one of our future works. In addition, if parents differ greatly from each other, the children will change greatly too, thereby bring about higher energy values. So how to make children effectively inherit parents' features is also our future work.

6. Conclusion

This study examines the effectiveness of PSAGC used in protein structure predictions. Through the experiments on three proteins, we found out that PSAGC is more powerful than conventional SA and PSA, it's able to discover lower energy conformations. Then we examine the energy transitions of the three proteins when using different CI. We find out that to bigger proteins, the shorter the CI be, the better the result be. Finally, we research the crossover methods of PSAGC, and the results of the experiment indicate the validity of crossover operator needs more investigations.

References

- [1] Simons K.T., Strauss C., Baker D., "Prospects for ab-initio protein structural genomics", J. Mol. Biol, Vol.306, pp. 1191-1199,2001.
- [2] Chance M.R., Bresnick A.R., Burley S.K.,et al, "Structural genomics: a pipeline for providing

- structures for the biologist", *Protein Sci.*, Vol.11, No.4, pp.723-738, Jan. 2002.
- [3] Frederic M.R., "The protein folding problem", *Scientific American*, Vol.264, NO.1, pp. 31-34, Jan. 1991.
 - [4] Anfinsen C.B., Haber E., Sela M. & F.H. White, "The kinetics of formation of native ribonucleases during oxidation of the reduced polypeptide chain", *Proc.Natl.Acad.Sci*, Vol. 47, NO 9, pp.1309-1314, Sep. 1961.
 - [5] Okamoto Y., "Protein Folding Problem as Studied by Monte Carlo Simulated Annealing", *Physical Properties Research*, Vol. 70, NO.6, pp. 719 - 742, 1998.
 - [6] WeiFeng Chen, KaiYang Li, Juan Liu, "The simulated annealing method applied to protein structure prediction", *Machine Learning and Cybernetics*, Vol. 5, pp. 2848 - 2852, 2004.
 - [7] Pedersen J.T., Moult J., "Genetic algorithm for protein structure prediction", *Current Opinion in Structural Biology*, Vol.6, No. 2, pp. 227 - 231, Apr 1996.
 - [8] Hiroyasu T., Miki M., Ogur M., "Parallel Simulated Annealing using Genetic Crossover", *Proceedings of IASTED2000 International Conference*, pp.139-144, 2000.
 - [9] Hiroyasu T., Miki M., Ogura M., and Okamoto Y., "Examination of Parallel Simulated Annealing using Genetic Crossover", *IPSJ JOURNAL*, Vol. 43, No. SIG 10(TOM 7), pp.70-79, 2002.
 - [10] Tanimura Y., Aoi K., Hiroyasu T., et al, "Implementation of Protein Tertiary Structure Prediction System with NetSolve", *Proceedings of Seventh International Conference on High Performance Computing and Grid in Asia Pacific Region*, pp. 320- 327, Jul.2004.
 - [11] Momany F.A., McGuire R.F., Burgess A.W., and Scheraga H.A., *J. Phys. Chem.*, Vol. 79, pp.2361 - 2381, 1975.
 - [12] Nemethy G., Pottle M.S., and Scheraga H.A., *J.Phys. Chem.*, Vol. 87, pp. 1883 - 1887, 1983.
 - [13] Sippl M.J., Nemethy G., and Scheraga H.A., *J.Phys. Chem.*, Vol. 88, pp. 6231 - 6233, 1984.